

The empirical recurrence for OEIS A251217: recovering a conjecture that is not written down

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Abstract

OEIS A251217 records “Empirical recurrence of order 62”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 128$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 62, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 62 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251217 is “Number of $(n+1) \times (6+1)$ 0..1 arrays with no 2×2 subblock having zero or two 1s.”. It has offset 1 and begins

631, 5569, 50912, 502971, 4907873, 48781920, 482274751, 4791965557, ...

The entry states:

Empirical recurrence of order 62 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:11:07, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 128$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 128$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 256$ exact terms of the model returns order 62 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{62} c_i a(n-i),$$

c_1	14	c_{17}	599803943307	c_{33}	−218380686071532085	c_{49}	−51928541426961232
c_2	100	c_{18}	−12296230808091	c_{34}	−82974313581710197	c_{50}	30441236882848704
c_3	−2046	c_{19}	1711941533195	c_{35}	389320099763519501	c_{51}	14709492440203264
c_4	−2051	c_{20}	83473592472856	c_{36}	58561944092853716	c_{52}	−9439759773564896
c_5	129809	c_{21}	−47674125113378	c_{37}	−563591899478647418	c_{53}	−3083087056948032
c_6	−153552	c_{22}	−450092212974811	c_{38}	15345515123317240	c_{54}	2140089848810304
c_7	−4756253	c_{23}	419246489196487	c_{39}	661722816417387670	c_{55}	458847957746944
c_8	11676192	c_{24}	1941203139150755	c_{40}	−111285646275597752	c_{56}	−342026276586112
c_9	112752778	c_{25}	−2445474935199405	c_{41}	−627963574450698980	c_{57}	−45418852652544
c_{10}	−392415267	c_{26}	−6711211209233756	c_{42}	177344846471204676	c_{58}	36219250451200
c_{11}	−1825478408	c_{27}	10675289735136813	c_{43}	478916022369018924	c_{59}	2653331189760
c_{12}	8371481014	c_{28}	18539825373363399	c_{44}	−180486384745614912	c_{60}	−2261365522432
c_{13}	20495642461	c_{29}	−36443592115080914	c_{45}	−291142080299007072	c_{61}	−68368564224
c_{14}	−126305891151	c_{30}	−40482480970315307	c_{46}	132924503138532656	c_{62}	62386864128
c_{15}	−153408995114	c_{31}	99321968763721236	c_{47}	139523482783336352		
c_{16}	1420945816061	c_{32}	68122294349844313	c_{48}	−73296043623097632		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 62, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $62 + 4$ terms removed returns order 62 as well, so $\deg q_{\min} = 62 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $62 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 62 that this sequence satisfies — and that family turns out to have exactly one member.

References

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